

5	(a) (i) frequency: number of oscillations <u>per</u> unit time of the source / of a point on the wave	M1 A1 [2]
	(ii) speed: speed at which energy is transferred / speed of wave <u>front</u>	B1 [1]
	(b) (i) does not transfer energy (along the wave)	B1 [1]
	(ii) position (along wave) where amplitude of vibration is a maximum	B1 [1]
	(iii) all three positions marked	B1 [1]
	(c) wavelength = $2 \times 17.8 = 35.6 \text{ cm}$	C1
	$v = f\lambda$	C1
	$v = 125 \times 0.356$	
	$= 44.5 \text{ m s}^{-1}$	C1
	$44.5^2 = 4.00 / m$	C1
	$m = 2.0 \times 10^{-3} \text{ kg m}^{-1}$	A1 [5]

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6 (a) either $P = VI$ and $V = IR$ or $P = V^2 / R$ C1